

Test and Reliability Exercise Sheet 1

Exercise 1

Let a sequential circuit C have the two primary inputs pi_1 and pi_2 , a primary output po_1 and four flip-flops $FF_1, \dots, FF_4$ as depicted in Figure 1. You can consider the circuit as a combinational core that is driven by the primary inputs as well as the secondary inputs ($si_1, \dots, si_4$) that come from the flip-flops. $so_1, \dots, so_4$ denote the secondary outputs of the circuit that at the same time serve as inputs for the corresponding flip-flop.

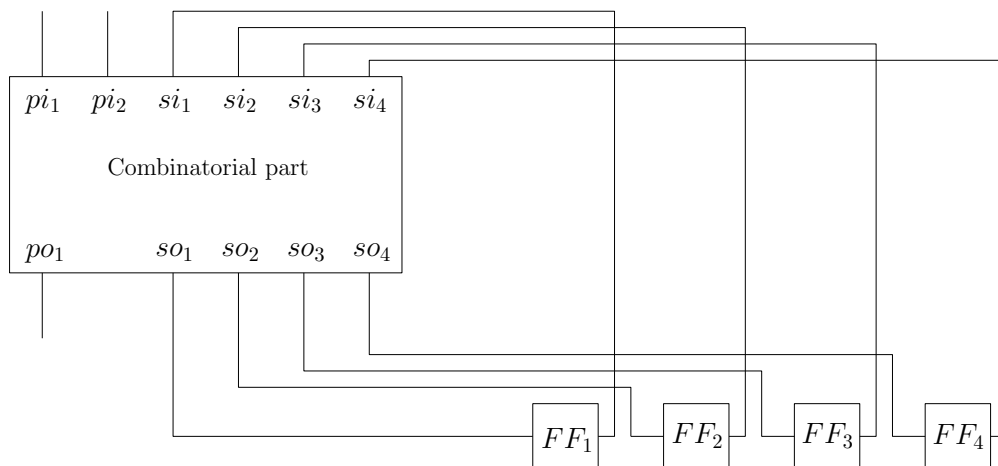


Figure 1: Circuit with two primary inputs, a single primary output and four flip-flops.

- Get familiar with this view on a sequential circuit.
- Consider the following two test vectors along with their expected responses (the primary inputs/outputs are given first separated by '|' from the secondary inputs/outputs):

- 1) 11|0011 $\rightarrow$ 0|0111
- 2) 01|0101 $\rightarrow$ 1|1100

Consider further the scan chain architecture depicted in Figure 2. Depending on the logic value of the `scan_enable` signal the scan chain is either transparent or in scan mode. In case `scan_enable` is set to '1', the muxes are set such that FF_1 captures the `scan_in` signal, while the remaining flip-flops capture the logic value of their respective preceding flip-flop. If `scan_enable` is set to '0' the scan chain is transparent and the circuit response is captured.

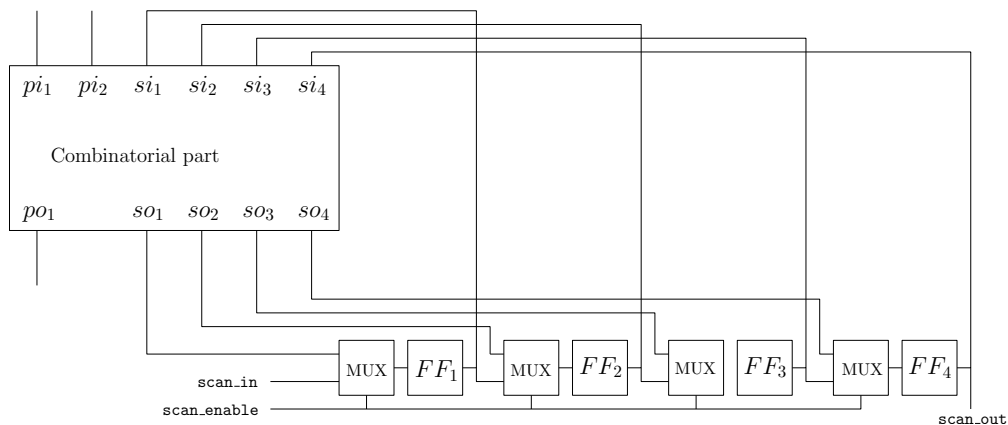


Figure 2: Extended circuit with a scan chain

Fill the following table with their corresponding signals such that the two test vectors are applied to the circuit ('X' identifies don't care signals) and write down the expected (i.e. fault free) output responses of the circuit. Assume that initially, each flip-flop is storing a logic '0'.

Clock cycle	pi_1	pi_2	scan_enable	scan_in	po_1	scan_out
1	X	X	1	1	X	0
2	X	X	1	1	X	0
3	X	X	1	0	X	0
4	X	X	1	0	X	1
5	1	1	0	X	0	1
6			1		X	1
7						
8						
9						
10						
11						
12						
13						
14						

Exercise 2

The circuit shown in Figure 3 is supposed to be tested. For this circuit the following test patterns have been chosen: 0101, 0111, 1010 and 1110.

- List all possible stuck-at faults in the circuit and perform fault collapsing. Make sure that the given test patterns detect all faults.
- Due to an accident during circuit manufacturing aluminum dust has been released. It is highly likely that aluminum particles up to a size of 12 nm will appear on metal layer 1. However, the

deposition of more than one particle on a single chip can be excluded. These particles are electrically conducting and can cause electrical shorts. The layout of metal layer 1 is shown in Figure 4 comprising the lines a to g. Identify all possible shorts.

c) Measurements reveal that the behavior of the shorts depends on the area of the corresponding polygons as follows:

- If one polygon is more than twice as large as the second one (in terms of area), the line of the larger polygon is stronger. This line is denoted as aggressor line, the weaker line as victim line. The logical value of the aggressor is always assigned to the victim.
- If this is not the case (i.e., the area of the polygons is roughly equal), the short behaves like a wired AND.

Determine the behavior of the identified shorts. Simulate the test patterns given above for these bridging faults. If some of these faults are not detected, give a test pattern which is able to detect them.

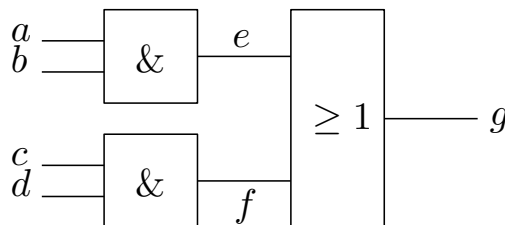


Figure 3: Circuit for the function $ab + cd$.

Exercise 3

- Consider the circuit from Figure 5 and briefly forget that a stuck-at-1-fault has been plotted already. Mark all the places in the circuit where a stuck-at fault can occur. So how many different single stuck-at fault have to be considered when testing the circuit?
- Try to generate a test for the stuck-at-1-fault (sa1) in the circuit depicted in Figure 5. Sensitize the fault first and then try to propagate the fault effect to the output. Please remember that you may need to sensitize several paths in order to propagate the fault effect to the output.
- Upon closer investigation of a failing circuit, it has been discovered that the stuck-at-1 fault in this particular circuit behaves like a slow-to-fall transition delay fault.
 - What is the relationship between a stuck-at fault and a transition fault?
 - Try to generate a test for a slow-to-fall transition delay fault at output of gate G1.

Deadline: Discussion on Thursday, May 07, 2026

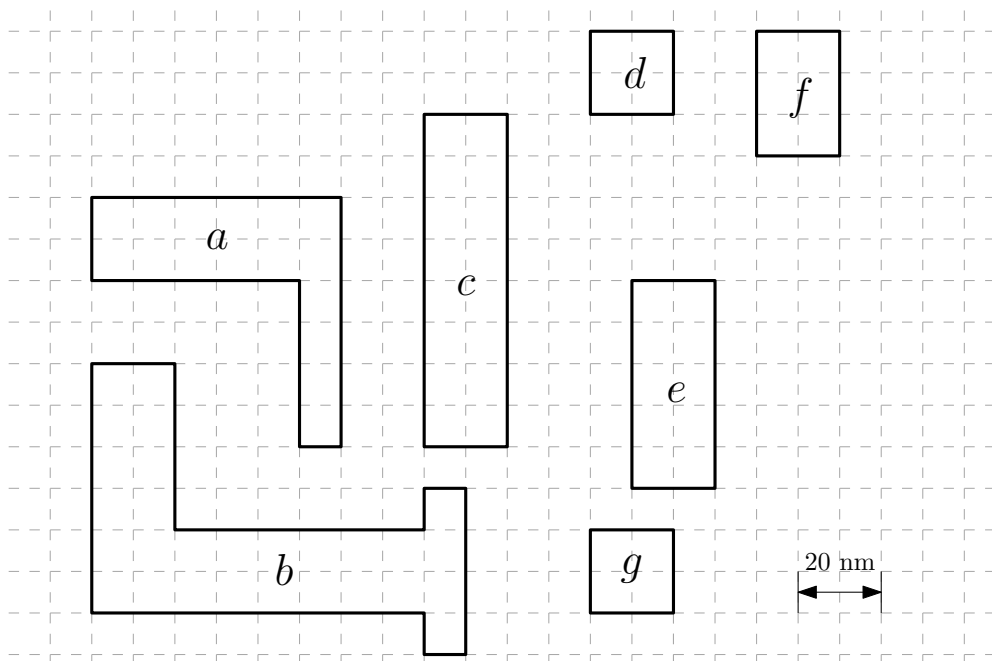


Figure 4: Layout of metal layer 1 of the circuit shown in Figure 3.

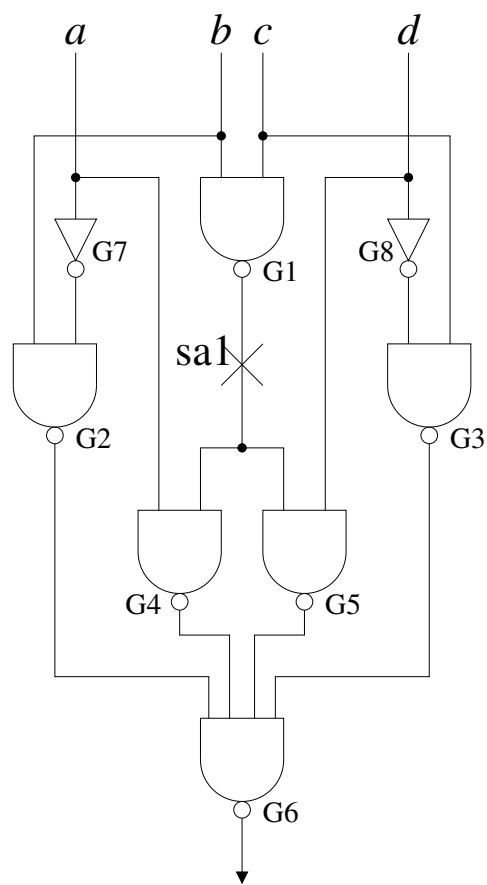


Figure 5: Faulty circuit